Retinal Disease Detection

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***Abstract*—Early detection of the disease can prevent permanent vision loss. However, accessing ophthalmologists and specialized diagnostic machinery is difficult in many rural and underserved parts of the world. We address this challenge by proposing an automated system for detecting retinal diseases, adopting a Computer Vision-based Deep Learning approach. It uses a fusion-based architecture between clip-vit-base-patch32 and DenseNet121, enabling the model to learn high-level semantic features and fine-grained patterns of retinal lesions. Fundus images are classified into five disease severity levels,from the APTOS 2019 Blindness Detection dataset. Then, it does binary classification, Data augmentation using random noise, rotation, and brightness adjustment will be done for improving robustness and generalizing. This solution can help reduce diagnosis delays and screening costs, providing accessible eye-care support, especially in the absence of medical specialists in various parts of the world, which would go along with global efforts in preventing avoidable blindness.**

***Index Terms*—Diabetic Retinopathy, Retinal Fundus Images, clip-vit-base-patch32, DenseNet121, Deep Learning, Computer Vision, Medical Image Classification, Multi-Class Disease Grading, APTOS 2019 Dataset, Feature Fusion, Healthcare Accessibility.**

I. Introduction

Human vision is one of the most important senses; damage to any part of the visual system may seriously affect daily life. Diabetic Retinopathy is considered one of the main causes of visual impairment in the world, which is a result of prolonged diabetes causing damage to the blood vessels of the retina. Early detection and timely treatment can prevent severe complications and permanent blindness. However, access to ophthalmologists and advanced imaging systems is often limited, especially in rural and underserved regions. The resulting delayed diagnosis increases the risk of irreversible vision loss.

Recent rapid development in the field of Computer Vision and Deep Learning has reached a point where automated screening systems can help medical professionals analyze retinal fundus images. Traditional CNNs are good at finding localized lesion patterns; however, their performance is often impacted by changes in image quality, with a weakness in terms of global contextual understanding. Our project focuses on developing a fusion-based deep learning approach by using CLIP for robust semantic feature extraction and DenseNet for fine-grained recognition of retinal diseases.

This system categorizes images into five different classes of DR severity using the APTOS 2019 Blindness Detection dataset, hence also reporting the risk of macular edema. The approach should improve diagnostic accuracy and make affordable eye-care support available even in remote areas.

II. Literature Survey

Early breakthroughs in automated diabetic retinopathy screening have shown us something remarkable: artificial intelligence can now spot signs of this sight-threatening condition just as well as experienced eye doctors. Gulshan and his team proved this by developing a deep learning system that analyzes photographs of the back of the eye, opening doors for AI to help screen more people in real-world medical settings.[11] What this really means is that computer-assisted diagnosis could help catch the disease earlier in far more patients—something especially valuable when screening large populations.

TABLE I
Benchmark of Retinal Disease Detection Models

Model	Accuracy / QWK
AlexNet [1]	97.9%
DenseNet-121 [2]	97.30%
EfficientNet-B3 [3]	∼0.821 QWK
VGG16 [4]	71.65%
ResNet-50 [5]	85.28%
Vision Transformer (ViT) [6]	98.79%
Custom CNN (CLAHE + ESRGAN) [7]	98.36%
clip-vit-base-patch32 + DenseNet121	**86.59%**
clip-vit-base-patch32 + DenseNet121	**∼ 0.7953 Cohen's Kappa**
clip-vit-base-patch32 + DenseNet121	**∼ 0.9237 QWK**

Getting the details right matters enormously when it comes to identifying the tiny abnormalities in the retina that signal trouble. That's why Pranay and colleagues refined a specialized neural network called DenseNet-121, making it better at spotting subtle warning signs like microaneurysms, small hemorrhages, and fatty deposits in the retina.[2] Medical research consistently reminds us that diabetic retinopathy gets worse over time, and accurately determining how advanced it is can make the difference between preserving someone's vision and losing it—which is why we desperately need dependable automated tools.[14]

The newest wave of research is getting even more sophisticated by teaching AI systems to understand both images and language together. Bianchi's team showed that clip-vit-base-patch32—a framework trained on millions of image-caption pairs from the internet—creates visual understanding that works surprisingly well even in specialized medical imaging.[12] A recent overview of how these vision-language models are being used in medicine revealed just how much potential CLIP-based systems have for helping doctors make better diagnostic decisions across all kinds of medical imaging scenarios.[14]

III. Methodology

Our proposed system brings together two powerful AI models—clip-vit-base-patch32 and DenseNet-121—to create a more accurate way of identifying diabetic retinopathy. Think of it as combining a bird's-eye view with a magnifying glass: one model captures the overall picture of what's happening in the retina, while the other zooms in on the tiny damaged areas that matter most. The entire process unfolds in four key steps: preparing the images, extracting features from two different perspectives, merging those insights, and finally making the diagnosis.

A. Preprocessing the Dataset

We work with the APTOS 2019 Blindness Detection dataset, a collection of real retinal photographs [10]. Before feeding these images into our system, we standardize them by resizing and normalizing. But here's the clever part: we also create slightly altered versions of each image—rotating them, adjusting brightness, cropping different sections, and adding a bit of artificial noise. Why? Because in the real world, photos come from different cameras, lighting conditions, and angles. By training on these variations, we're essentially teaching the model to handle whatever it might encounter in an actual clinic.

B. Dual Feature Extraction

Instead of relying on just one AI model, we use two working in parallel on the same image:

- **clip-vit-base-patch32** acts as the "big picture" thinker. It has been trained on millions of images and captions from across the internet, allowing it to understand broad visual patterns and global context [1].
- **DenseNet-121** serves as the "detail detective"—its architecture is specially designed to spot specific retinal abnormalities such as tiny blood vessel leaks or hemorrhages[2].

Each model produces its own set of insights—a feature vector that captures what it "sees" in the image.

Bringing it all together, we take the insights from both models and merge them into a single, unified understanding, enabling the system to leverage both global semantic and fine-grained structural information.

This combined view passes through what we call a **projection layer**—essentially a translator that helps the two different "languages" work together smoothly.

The result is a system that can simultaneously consider:

- The overall health and structure of the retina (from **clip-vit-base-patch32**), and
- The specific warning signs and lesions (from **DenseNet**).

C. Classification Performance

TABLE II
Overall Classification Performance Metrics

Metric	Score
Overall Accuracy	85.45%
Macro F1-Score	0.73
Weighted F1-Score	0.85
Quadratic Weighted Kappa (QWK)	0.92

The QWK score of 0.92 indicates a high level of agreement between model predictions and expert-labelled ground truth, confirming strong reliability for clinical-level disease grading.

D. Class-wise Performance

TABLE III
Class-wise Performance Metrics for DR Severity Levels

Class (DR Level)	Precision	Recall	F1-Score
0 — No DR	0.9682	0.9861	0.9771
1 — Mild	0.6667	0.5778	0.6190
2 — Moderate	0.7429	0.8595	0.7969
3 — Severe	0.6667	0.4348	0.5263
4 — Proliferative	0.8846	0.6571	0.7541

The model performs highly accurately for **No DR** and **Moderate DR**, which represent the most common clinical categories. However, **Mild** and **Severe** cases show lower recall due to class imbalance and subtle lesion variations. These classes can be improved by increasing sample diversity or adopting class-balanced loss functions.

E. Interpretation

The fusion approach successfully strengthens disease discrimination by combining:

- **DenseNet** lesion-level structural features, and
- **clip-vit-base-patch32** global contextual embeddings.

This integration leads to high grading reliability and strong agreement with clinical assessment, demonstrating the effectiveness of the dual-feature extraction strategy.

F. Confusion Matrix

- The model rarely misclassifies healthy images as diseased.
- Most confusion occurs between **Moderate** $\leftrightarrow$ **Severe** and **Moderate** $\leftrightarrow$ **Proliferative** stages, which is expected since these stages share similar retinal lesion patterns.

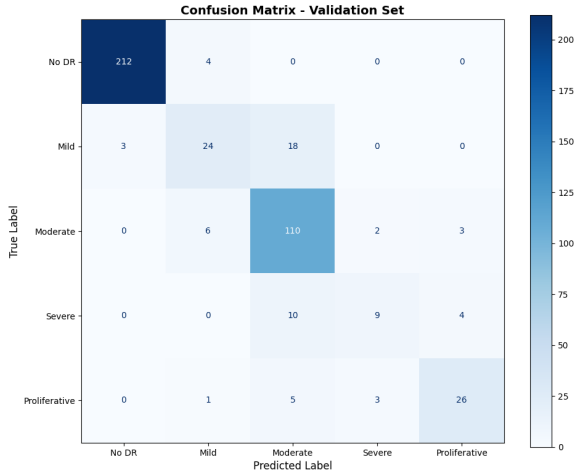


Fig. 1. Confusion Matrix

IV. Grad-CAM Visualizations

Grad-CAM (Gradient-weighted Class Activation Mapping) was utilized to highlight retinal regions that most strongly influenced the model's predictions.

This visualization enhances the interpretability of the proposed system and supports clinical trust in the diagnostic decisions made by the **clip-vit-base-patch32 + DenseNet Fusion Model**.

The system is trained carefully in two stages:

1) **Stage 1:** The decision-making layers are trained while keeping the CLIP and DenseNet models frozen—similar to learning how to interpret what experts are saying before becoming an expert yourself.
2) **Stage 2:** The entire network is fine-tuned, gradually allowing DenseNet (and selectively parts of CLIP) to adjust and specialize for this specific diagnostic task.

Explainability

One of the most important features of the proposed system is its explainability. The **Grad-CAM** (Gradient-weighted Class Activation Mapping) technique is applied on the DenseNet pathway to generate heatmaps that highlight the specific retinal regions influencing the diagnosis [15].

V. Results

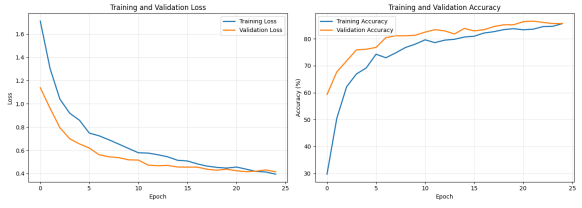


Fig. 2. Loss and Accuracy graphs of proposed model .

The performance of the proposed **clip-vit-base-patch32 + DenseNet121 Fusion Model** was evaluated on the APTOS 2019 test split using several performance metrics, including

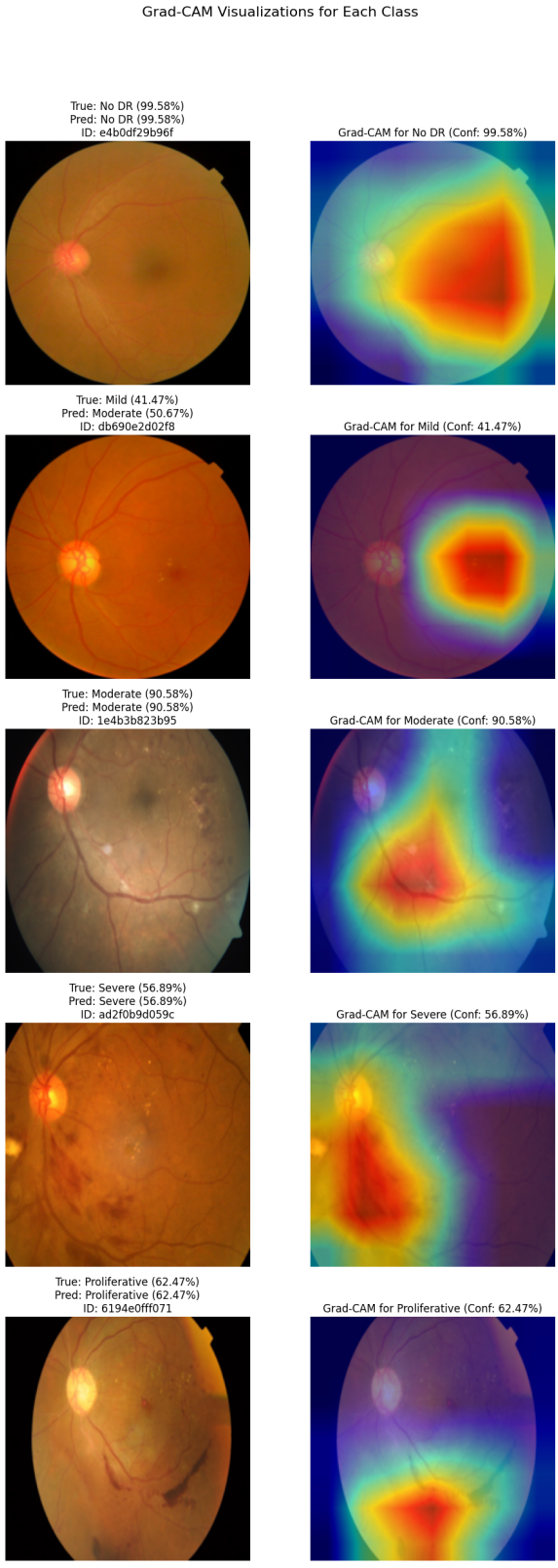


Fig. 3. Grad-CAM highlighting hemorrhages and vascular abnormalities.

accuracy, precision, recall, F1-score, confusion matrix, and Quadratic Weighted Kappa (QWK).

QWK is particularly significant in medical grading tasks as it penalizes misclassifications based on the difference in severity between the predicted and ground truth grades.

VI. ACKNOWLEDGMENT

In this project, we developed an automated system to detect and grade diabetic retinopathy by integrating two complementary AI approaches—**clip-vit-base-patch32** and **DenseNet-121**—each contributing unique strengths.

DenseNet-121 excels at identifying the fine-grained retinal abnormalities such as microaneurysms, small hemorrhages, and fatty deposits that ophthalmologists typically examine during eye assessments. Meanwhile, clip-vit-base-patch32 provides a broader contextual understanding of the overall retinal image and effectively manages challenges such as poor illumination or variations in image capture quality.

By fusing these two perspectives through a dedicated combination (projection) layer, the system achieves enhanced accuracy and consistency across all stages of the disease. Data augmentation strategies were employed to simulate real-world variability, improving robustness and preventing overfitting.

The resulting model can reliably assess diabetic retinopathy severity and indicate the presence of macular edema—the swelling that can impair central vision. This system has the potential to be transformative in **rural and underserved regions** where access to ophthalmologists is limited.

Rather than replacing clinicians, this system extends their diagnostic reach—facilitating early detection, cost-effective screening, and prevention of avoidable blindness. Ultimately, this work represents a step toward making eye care more accessible and equitable worldwide.

VII. FUTURE WORK

Although the current model achieves strong performance, several directions remain for further enhancement:

1) **Dataset Expansion:** Incorporate larger and more diverse retinal datasets to improve generalization across populations and imaging devices.
2) **Lightweight Deployment:** Compress and optimize the model for deployment on edge or mobile platforms, enabling real-time screening in field clinics.
3) **Clinical Integration:** Develop a user-friendly diagnostic interface for healthcare professionals, including automated report generation and referral recommendations.
4) **Multi-Disease Detection:** Extend detection capability to other retinal diseases such as glaucoma, cataract, and age-related macular degeneration.
5) **Enhanced Explainability:** Explore advanced interpretability methods beyond Grad-CAM—such as attention rollouts—to improve trust and transparency in clinical usage.

These improvements will help transition the proposed system from research to real-world clinical application, making early eye disease detection more accessible and reliable.

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APPENDIX: AUTHOR CONTRIBUTIONS (BASED ON CREDIT TAXONOMY)

TABLE IV
AUTHOR CONTRIBUTION SUMMARY

CRediT Role	Arun Kumar	Charan Teja
Methodology	Lead	Support
Software (Coding & Implementation)	Support	Lead
Data Curation	Support	Lead
Writing – Review & Editing	Lead	Support
Project Administration	Support	Lead